

# Home Bias and International Risk Sharing

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## **Abstract**

This paper investigates the relationship between equity home bias and international risk sharing. The analysis is based on panel-data regressions using OECD national accounts data and foreign asset holdings.

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# 1 Introduction

In their 2007 Paper Sørensen et al. argue that risk sharing and home bias, are two economic patterns that were analysed separately before but "may be manifestations of the same underlying behavior".

After replicating the main results of the paper, and extending it by time and sample size<sup>1</sup> we shift our focus the eurozone as an interesting case study, to further investigate how home bias and international risk sharing develops. European Integration during the original papers sample period (1993–2003) and afterwards presents a fascinating opportunity to investigate, whether a decline in the exchange rate risk and transaction costs, two main reasons for Home Bias, is actually associated with a declining Home Bias and increasing risk sharing. Naive data analysis finds that the Euroarea countries became more diversified during their integration, but not within themselves, but rather with countries outside the Eurozone. This highly puzzling result motivated us to analyse the subject further, finding that Financial Centers inside Europe are the main drivers of this trend and present a major unsolved challenge to the original Sørensen et al. paper. Hence, rather than doing a single case study, we shifted our papers focus to analyse this shortcoming more throughoutly.

## 2 Panel regressions

### 2.1 Theoretical outline

The authors use a panel regression setting to analyse their central question - whether a lower home bias is associated with a higher level of income risk sharing and consumption risk sharing. To do so, they regress the log growth rates of country specific Gross National Income (GNI) or Consumption (C) relative to the full sample's log growth rate of GNI or Consumption on the country specific log Gross National Product (GDP) growth rate, relative to the full samples GDP growth rate. The baseline regressions are

$$\Delta \log GNI_{it} - \Delta \log GNI_t = \alpha_i + \kappa (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it} \quad (1)$$

and

$$\Delta \log C_{it} - \Delta \log C_t = \alpha_i + \eta (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it}. \quad (2)$$

As in the cross sectional regressions, one minus the respective risk sharing coefficients  $\kappa$  and  $\eta$  measures the average amount of risk sharing of a country.

To include the equity home bias into this regression, the authors apply an interaction term for the equity home bias of a country, relative to the average equity home bias in

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<sup>1</sup>As much as possible, constrained by data availability.

the sample and an interaction term for a potential time trend.

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(EHB_{it} - \overline{EHB}_t) \quad (3)$$

and

$$\eta = \eta_0 + \eta_1(t - \bar{t}) + \eta_2(EHB_{it} - \overline{EHB}_t). \quad (4)$$

Once again it is most intuitive to interpret the average risk sharing coefficient of a country:

$$AverageIncomeRiskSharing = 1 - \kappa_0 - \kappa_1(t - \bar{t}) - \kappa_2(EHB_{it} - \overline{EHB}_t) \quad (5)$$

and

$$AverageConsumptionRiskSharing = 1 - \eta_0 - \eta_1(t - \bar{t}) - \eta_2(EHB_{it} - \overline{EHB}_t). \quad (6)$$

For the hypothesis of a potential co-movement between the equity home bias and risk sharing we must focus on the third coefficients  $-\kappa_2$  and  $-\eta_2$ , implying that an increase in the equity home bias by 0.1 increases income risk sharing by  $-\kappa_2$  and consumption risk sharing by  $-\eta_2$ .

All panel regressions are run as two-stage feasible Ordinary Least Squares Estimations. In the second stage regression each country is attributed a weight relative to 1 over its error terms standard deviation in the first stage regression.<sup>2</sup>

## 2.2 Empirical results

When reporting the results of the panel regressions, we follow the authors standard and report the easier to interpret coefficients  $1 - \kappa_0/1 - \eta_0$ ,  $-\kappa_1/1 - \eta_1$  and  $-\kappa_2/1 - \eta_2$  multiplied by 100 to interpret them as percentage effects. Reassuringly, our panel regression results mainly follow the key results of the Sorensen et. al (HIER LINK EINFÜGEN) paper, with a statistically non-different to zero estimate for the time trend and a highly significant coefficient for the equity home bias on both income and consumption risk sharing. An increase in the equity home bias by 0.1 is now associated with an increase in income risk sharing by roughly 13 percent and consumption risk sharing by roughly 20 percent. Given that our point estimates for the coefficient are larger than 100 and notably higher than the point estimates found in the Sorensen et al (HIER LINK EINFÜGEN)

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<sup>2</sup>Since this is weakly documented in the original paper, we compared the first and second stage regression results and conclude that the differences have no impact on the direction and significance of coefficients.

paper, we believe that data revisions and the choice of the base year (AS EXPLAINED ABOVE) play an important role. Further, we rely on an interpretation offered by the authors themselves, believing that: "a large decline in home bias is not likely to lead to more than 100 percent risk sharing [...] and we interpret the coefficient to imply that declining home bias has been associated with strongly increasing [...] risk sharing, even if the actual value is likely to not be valid for very large changes in home bias." Further, our coefficient for the average income risk sharing is highly significant, while it is not in the original paper, something that we blame on data revisions and the choice of the base year.

In table 5, we follow the papers approach and run similar income risk sharing and consumption risk sharing regressions on terms of certain foreign asset holdings over GDP.<sup>3</sup> We treat these regressions merely as an robustness check, and find positive  $-\kappa_2$ ,  $-\eta_2$  coefficients for all forms of asset holdings, and statistically non significant estimates for the time trend.

Even though we find partly differing point estimates, due to data revisions or the choice of the different base year, our findings is mainly in line with the authors finding for the same country and time sample. Therefore, we conclude that the main findings of the Sorensen et al (HIER PAPER EINFÜGEN) panel regression can be replicated.

### 3 Extentions

#### 3.1 First Extention: Similar Country Sample 1987–2017

We aim to analyse how the 22 countries of the original analysis have developed during a by 20 years extended time period from 1987 to 2017. This extention is particularly interesting as the original period of analysis of 11 years is rather short.

We find that the results of the authors analysis and our replication largely hold when extending the time period by 20 years. A 0.1 decrease of the equity home bias is now associated with a roughly 3 percent increase in income risk sharing (HIER TABELLE VERLINKEN) and a 8 percent increase in consumption risk sharing.

The estimate for the time trend remains statistically non-significant and the average risk sharing coefficient for income and consumption risk sharing decreases notably with the coefficient for average income risk sharing becoming statistically non significant whilst the coefficient for average consumption risk sharing remains highly significant. Given that the average income risk sharing coefficient was non-significant in the Sorensen et al (HIER LINK EINFÜGEN) paper in the original sample, we believe that the change may potentially come from data differences and the choice of base years.

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<sup>3</sup>Note that a positive sign of  $-\kappa_2 / -\eta_2$  is now associated with a higher level of risk sharing.

Looking at table 5 we can see that the same results hold for the foreign assets over GDP regressions, giving us confidence to conclude that the negative relationship between Home Bias and Income and Consumption Risk sharing holds for the extended period of 30 years between 1987–2017.

### **3.2 Second Extention: Extended Country Sample 1987–2017**

Further, we aim do analyse how the statistical relationship holds between 1987 and 2017 for a larger sample of OECD countries. With respect to the panel regressions, extending the sample period beyond 29 countries was not possible due to lacking data coverage either for GNI or for the Equity Home Bias measures. A starting point for future research would defenitely be a larger extention for economies not included in our dataset, especially given the fact that the relative importance of non OECD economies has increased over the past two decades since the paper was published. The panel-regression results of the (slightly) extended sample between 1987 and 2017 confirm the same results as shown in the two regressions before. An decrease in the Equity Home Bias by 0.1 is associated with an increase of income risk sharing by roughly 3 percent and an increase in consumption risk sharing by roughly 8 percent. again, the panel regressions for foreign Asset holdings relative to GDP confirm this estimation, with highly significant estimations for the asset holdings themselves, and non significant estimations for the time trend. Again, the average risk sharing coefficient for income is non significant in both specifications, while it is highly significant for consumption in both specifications.

## **References**

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